Part 1 Brief · Foundation

mcp-ops-server · 17 Oct 2026 · PR #1 · Status: CI green on push and PR, ready to merge

WHAT WAS BUILT

- **A running MCP server** with one `health` tool, reachable over Streamable HTTP at `/mcp`. Verified by a real client call, not just unit tests: it returned `{status: ok, version: 0.1.0}`.
- **CI on every push and PR:** lint (ruff), format check, tests (pytest), and a secret scan (gitleaks) over full git history.
- **The project's operating system:** `comms/PLAN.md` (architecture and 6 parts), `STATUS.md`, `AGENTS.md` (rules any coding agent reads), `FAILURES.md`, MIT license.

HOW IT WORKS

File	Role
mcp_ops_server/server.py	build_server() creates an MCPServer and registers tools. A factory function, so every test gets a fresh server with no shared state.
mcp_ops_server/__main__.py	python -m mcp_ops_server starts it; host and port come from env vars (MCP_OPS_HOST, MCP_OPS_PORT), never hard-coded.
tests/test_health.py	Lists tools and calls health through the SDK's own call_tool, so the test exercises real MCP schema generation.
.github/workflows/ci.yml	Three parallel jobs: lint, test, secrets.

WHY IT WAS DONE THIS WAY

- **MCP SDK v2, pinned.** Most tutorials use `FastMCP` (v1). In v2 it is `MCPServer`. Pinning `mcp>=2.3,<3` stops silent breakage from either direction.
- **TypedDict return type.** With a plain `dict`, the SDK sends only text, so agents get no machine-readable output schema. A TypedDict makes the SDK publish an output schema and return `structured_content`. This matters in Part 4, where evals check tool outputs field by field.
- **Secrets scanned from commit one.** The repo goes public later, and history goes with it. Scanning early means nothing ever needs scrubbing.
- **Synthetic data only**, written into the agent rules as a hard rule, not a convention.

WHAT WENT WRONG (LOGGED IN FAILURES.MD)

- **Agent delegation over GitHub failed.** @codex got no response; @cursor was gated behind a paid add-on. New workflow: Claude implements, Codex independently reviews each part.
- **Secret scan failed only on PRs.** My first fix scoped it to push. Codex's review (P1) caught that fork PRs would then skip scanning. Final fix: gitleaks CLI on push and PR, read-only token.
- **Codex review: 3 findings, all fixed.** The fork-PR gap above; tests never started the real server (added an HTTP integration test that fails fast if startup breaks); the brief generator's dependency was undeclared (added a `docs` extra).

VERIFY IT YOURSELF

```
pip install -e ".[dev]"
ruff check . && pytest -q          # 3 passed (incl. live HTTP test)
python -m mcp_ops_server           # serves http://127.0.0.1:8000/mcp
```

INTERVIEW TALKING POINT

"I treated tool output schemas as an API contract from day one: the health tool returns a typed structure, so agents and evals can validate fields instead of parsing text."

NEXT: PART 2, CORE SERVER

Synthetic clinic data in SQLite, five tools (search_providers, list_open_slots, book_appointment, cancel_appointment, get_referral_status), unit tests, then a Codex review. **Your action:** merge PR #1 on GitHub, then reply "go".